1. Size comparison of P2PKH and P2SH-P2WPKH transactions
From the decoded transactions:

P2PKH (Legacy) transaction
Size: 191 bytes
vsize: 191 vbytes
Weight: 764

P2SH-P2WPKH (SegWit) transaction
Size: 215 bytes
vsize: 134 vbytes
Weight: 533

At first, the SegWit transaction appears larger in raw bytes (215 vs 191). However, what actually matters in Bitcoin is vsize (virtual size) and weight, because transaction fees are calculated based on these values.

So the effective size comparison is:

Transaction Type	Raw Size	Virtual Size	Weight
P2PKH	191 bytes	191 vbytes	764
P2SH-P2WPKH	215 bytes	134 vbytes	533

Thus, SegWit transactions are effectively smaller and cheaper even though their raw byte size is slightly larger.

2. Script structure comparison

P2PKH (Legacy)
In P2PKH transactions, the unlocking data is placed directly in the input script (scriptSig).

scriptSig (unlocking script):
<signature> <public key>

scriptPubKey (locking script):
OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG

Execution stack:
<signature> <pubkey>
OP_DUP
OP_HASH160
<pubKeyHash>
OP_EQUALVERIFY
OP_CHECKSIG

So in legacy transactions:
Signature and public key are stored inside the transaction input
Everything counts fully toward the transaction size.

P2SH-P2WPKH (SegWit)
SegWit separates the unlocking data from the transaction.

scriptSig:
0 <pubKeyHash>
This is just a small redeem script.

Witness data:
<signature>

```
<public key>

scriptPubKey:
OP_HASH160 <scriptHash> OP_EQUAL

Validation steps:
Check that
HASH160(0 <pubKeyHash>) == scriptHash
If correct, execute the underlying SegWit script:
<signature> <pubkey>
OP_DUP OP_HASH160 <pubKeyHash> OP_EQUALVERIFY OP_CHECKSIG
```

Thus the actual signature verification is the same as P2PKH, but the signature and public key are stored in the witness section instead of scriptSig.

3. Why SegWit transactions are smaller

SegWit reduces the effective transaction size because of weight calculation.
Bitcoin uses the following rule:
Weight = (Base transaction data x 4) + Witness data

Base data --> version, inputs, outputs, scripts
Witness data --> signatures and public keys

Witness data is counted at 1x weight, while base data counts at 4x weight. Since signatures are the largest part of a transaction, moving them to the witness section significantly reduces the effective size (vbytes).

This is why:
P2PKH vsize = 191
SegWit vsize = 134

Even though the raw byte count is slightly larger.

4. Benefits of SegWit transactions

SegWit provides several advantages:

1. Lower transaction fees
Since the effective size (vbytes) is smaller, users pay lower fees for the same transaction.

2. Increased block capacity
Because witness data is discounted, more transactions can fit into a block. This increases the effective throughput of the Bitcoin network.

3. Fixes transaction malleability
Before SegWit, the transaction ID could be modified by changing the signature encoding. SegWit separates signatures from the transaction hash calculation, which prevents transaction malleability attacks.

4. Enables advanced features
SegWit made it possible to build newer Bitcoin technologies such as Lightning Network, Taproot, more efficient smart contract constructions